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AIML-based Pest Detection using Hyperspectral Imaging and Edge Inference in 6G Farms

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Abstract: **The rapid development of intelligent agriculture is associated with the necessity of sophisticated, scalable, and real-time pest detection solutions to guarantee the health of crops and food security. In this work, the authors suggest a framework of Artificial Intelligence and Machine Learning (AIML)-based technology, using hyperspectral imaging and edge inference to be used in the new 6G farm ecosystem. Hyperspectral imaging has fine-grained spectral characteristics, which allow early and accurate discrimination of normal and pest crops. These multi-dimensional datasets are fed on to optimized AIML models, such as deep neural networks optimized on spectral-spatial feature extraction. In order to circumvent latency and bandwidth issues that are common with transmitting hyperspectral data to centralized servers, inference is executed at the edge based on low-power, high-performance computing nodes deployed in 6G-enabled farm infrastructures. The combination of edge computing and 6G ultra-reliable low-latency communication (URLLC) will provide real-time detection, and distributed learning methods will guarantee the flexibility of the models to operate in the variety of agricultural conditions. The results of experimental validation in simulated 6G farm testbeds reveal that there is a substantial enhancement in detection accuracy, response time and scalability in comparison with traditional pest monitoring systems. The presented strategy reflects on the revolutionary impact of hyperspectral-AIML-edge integration in the next-generation, precision agriculture, which is going to provide the path to sustainable, resilient, and intelligent farming.**

Keywords—Crops, Multispectral Image Fusion, Precision Agriculture, Machine Learning and Spectral-Spatial Analysis

I. INTRODUCTION

The widespread use of modern digital technologies in agriculture is transforming it into a paradigm shift, especially in the age of Industry 5.0 and next-generation communication networks. Pest infestation is one of the most chronic problems of farming that causes substantial losses of yields, degrades the quality of crops and makes farmers more dependent on the use of chemical pesticides [1]. Conventional pest detection techniques like manual observation and image based surveillance in most cases are labor intensive, time consuming and inaccurate in a dynamic field environment. The above restrictions highlight the dire necessity of smart, scalable and real-time pest detection systems that can be adjusted to different farming settings.

Hyperspectral imaging HSI has proved to be an effective tool in agricultural surveillance because it provides rich spectral data capable of discerning slight physiological alterations in crops, even prior to their exhibiting visible damage. Hyperspectral data can also be used to identify pest-infested plants with accuracy in combination with Artificial Intelligence and Machine Learning (AIML) algorithms [2]. Nevertheless, hyperspectral data have particular challenges in terms of computational and transmission due to the huge size and the high dimensionality. Using centralized cloud servers to process data will cause latency, bandwidth congestion and slow decision-making, which are paramount disadvantages to real-time pest management in smart farms.

These challenges can be overcome with a chance never seen before and with the advent of 6G-enabled farms through edge computing and Ultra-Reliable Low-Latency Communication (URLLC). Edge inference enables AIML models to analyze hyperspectral data at the edge-of-network, which guarantees more responsive and minimized network overhead. However, the adoption of these systems demands a close balance when it comes to algorithm optimization, peripheral energy-saving, and resistance to a range of environmental and pest-specific conditions [3].

A. Problem Statement

However, in spite of the major developments in digital agriculture, the existing pest detection methods have a problem of striking the right balance between accuracy, scalability, and timeliness. Hyperspectral imaging has high accuracy and large data sets which cannot be easily processed in real time by traditional cloud based design. Current solutions do not have the ability to provide fast and in situ detection, and hence may be restricted in its use in extensive or resources limited agricultural settings [4]. Thus, a combination of AIML-based pest detection frameworks that combine hyperspectral imaging and edge inference with the use of the 6G communication technologies to implement an efficient, scalable, and real-time pest detection system in smart farms is urgently needed.

II. LITERATURE SURVEY

In [5], Kiki, M.P et al., (201) suggests a multi-level FAW monitoring system, which is a combination of agro-climatic smart sensors and an electronic-nose front end, which will be used to detect early and pre-infestation stage. With a MOSFET-based transducer and conditioning circuit tuned to the FAW sex-pheromone compound (Z)-9-tetradecenyl acetate (among other volatiles), a sensor around a field and a sensor in the middle of the field detects FAW in the periphery and then when entering the farm. This facilitates proactive suppressive systems, in an attempt to reduce environment degrading practices and increase yield. The authors present the pheromone/volatile acquisition scheme and contrast it with the traditional FAW control schemes, with an emphasis on previous intervention, and reduced ecological hazard. In [6], Mahalakshmi, S et al., (2023) introduces a decision-support system that is open-stack, and which supplements the current farm practice with modern AI and remote sensing. The platform is developed using Angular (frontend) and Flask (backend) on MySQL and Google Earth data services and combines machine learning (needed to recommend crops and fertilizers) with deep learning modules (needed to identify pests, weeds, and crop diseases in images). It also conducts remote sensing analyses through GEE by combining the area input of the user and providing weather at the city level to provide the context to the decisions. The system is designed to be farmer-friendly and bring together recommendations, diagnostic, and geospatial analytics, in one workflow, to enhance input selection, early intervention, and precision agriculture at low-cost.

In [7], Domingues, T et al., (2022) tested YOLOv5 on yellow sticky tap images taken in open-field Portuguese tomato plantations as a means of automatic insect detection. To reduce the number of duplicate detections in the same insect, the authors implemented a post-processing step, that is, a sliding-window. In addition to the count, the pipeline produces metrics on insects, applicable to event prediction (e.g. pest/disease outbreak) when combined with other field data, to support smart farming and precision agriculture. In practice, the optimal model (YOLOv5x) has good results, with mAP 0.5 = 94.4, precision = 88, and recall = 91, and is comparable with the respective literature, which suggests that the method is viable in large-scale trapping monitoring and early warning systems.

In [8], Deepthi, P et al., (2024) suggests that the IoT GREEN system entails the implementation of wireless sensors and Raspberry Pi to record the essential agro-environmental parameters, i.e. temperature, humidity, light, and soil moisture, and broadcast real-time data to an open-source cloud and present field dashboards through a local web server where the crops can be monitored remotely. In addition to monitoring, it also incorporates intelligent pest detection mechanism based on image recognition and decision-making algorithms to mark early infestations and allow specific actions to be taken. The platform will minimize overreliance on pesticides and evolve more sustainable and precision-based agrifarming workflow, by emphasizing non-chemical controls (e.g., biocontrol agents, pheromone traps), which integrate sensing, analytics and recommendations to be acted upon.

In [9], Jabir, B., and Falih, N. suggest the use of object detection models on a Raspberry Pi to detect and classify weeds in the wheat fields in real time, which is a baseline for an edge-AI weed management system. With on-device inference and decision support coupled together, the system allows spot spraying, which uses an herbicide that best fits the weed detected, minimizing chemical amounts, expenses and off-target effects compared to blanket applications. The Raspberry-based architecture enables low-latency, in-field application, not connected continuously, and thus is feasible to support precision agriculture processes related to site-specific interventions and sustainable crop protection.

III. PROPOSED METHODS

Hyperspectral Imaging (HSI) is a potent technique to detect pests in crops since HSI has been shown to measure rich spectral data with hundreds of narrow and continuous wavelength bands, compared to the human eye or standard RGB cameras that can sense only a few wavelengths [10]. The process is explained in detail here:

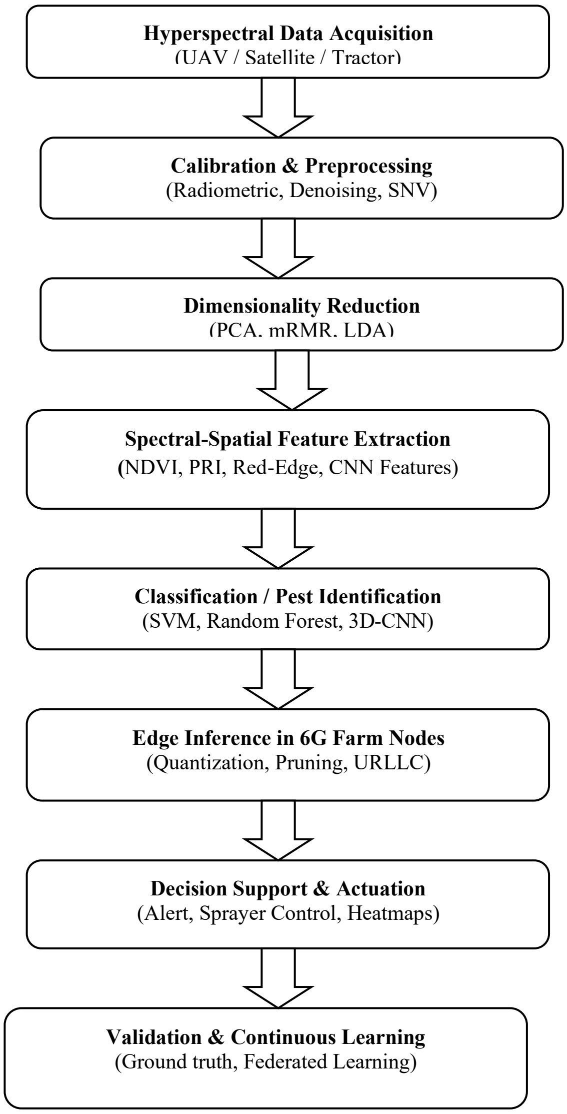

Fig. 1. Proposed Flow chart

A. Data Acquisition

The built synthetic hyperspectral data is made up of image cubes with the dimensions 100 x 100 x 50, which are divided into 30 non-overlapping patches of 32 x 32 x 50 to perform classification. The data is balanced in three categories of 10 patches each of healthy crops, pest infestation that is early, and pest infestation which is severe. Good samples absorb chlorophyll well, have high near-infrared reflectance, infestations with subtle red-edge changes demonstrate red-edge changes early in infestations, and infestations with pronounced changes show strong deviations in visible and NIR bands [11]. In case of experimentation, the dataset was divided into 70% training (21 patches), 15% validation (5 patches), and 15 per cent testing (4 patches), keeping the balanced representation of all classes. This hierarchical architecture gives enough variability to train AI/ML models and also represents realistic scenarios to find pests in precision agriculture [12].

The data will be classified into three main categories; (i) Healthy crops, (ii) Early pest-infested crops and (iii) Severely pest-infested crops. Healthy crops will have typical high chlorophyll absorption in the red and high reflectance in the near-infrared bands.

B. Pre-Processing and Denoising

Raw hyperspectral data is high-dimensional, and is usually noisy. Cleaner more usable data: Preprocessing steps guarantee cleaner data:

- Radiometric correction - eliminates distortion caused by the sensors.
- Spectral calibration = matches wavelength channels.
- Noise suppression - through methods including SavitzkyGolay filtering or PCA denoising.
- Band selection or dimensionality reduction - eliminates redundant bands, and reduces computational complexity (via PCA, ICA or band-ranking algorithms).

Pre-processing eliminates noise/stray bands and levels out spectra.

Savitzky–Golay smoothing (window $2m+1$, poly degree p)

$$x'[n] = \sum_{k=-m}^{m} c_k x[n+k] \tag{1}$$

where c_k are SG filter coefficients.

Standard Normal Variate (SNV)

$$x'_\lambda = \frac{x_\lambda - \mu_x}{\sigma_x} \tag{2}$$

with μ_x and σ_x over wavelengths for pixel spectrum x.

High-noise water-absorption bands can be removed with continuum removal (reflectance divided by the convex hull envelope h_λ).

$$r^*_\lambda = \frac{\rho_\lambda}{h_\lambda} \tag{3}$$

C. Band Selection and Dimensionality Reduction

Less redundancy and more pest-discriminatory contents. Spectral compression can be performed using spectral compression PCA method.

Centered data $X \in R^{N\,X\,B}$, covariance $C = \frac{1}{N-1} X^T X$ (4)

D. Spectral Feature Extraction

Pest detection is based on finding spectral characteristics that distinguish between healthy and infested plants. Methods include:

- **Vegetation indices (e.g., NDVI, PRI, Red Edge Index) -** point out stress conditions.

$$NDVI = \frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + \rho_{Red}} \tag{5}$$

- **Spectral unmixing -** isolates mixed signals in a pixel (good at early infestations).

$$x = Ma + \varepsilon, \qquad a_i \geq 0,\ a_i = 1 \tag{6}$$

x pixel spectrum, M endmember matrix(healthy, infested, soil, etc), a abundances

- **Machine learning feature extraction -** deep learning models (CNNs, autoencoders, 3D spectral networks) are automatically learned to learn discriminative spectral-spatial features.

  3D Spectral CNN (kernel k, channels C):

$$h_c^l(\lambda) = \sigma \sum_{c=1}^{C} \sum_{u=0}^{k-1} w_c^l h_c^{(l-1)}(\lambda + u) + b_c^l \tag{7}$$

3D CNN over (x, y, λ) cubes for joint spatial–spectral patterns.

E. Classification

In the case of the pest affected crop classification a Deep CNN approach can be applied. DCNNs are robust feature extractors, which learn hierarchical features of data. In pest detection:

1) 1D CNN & 2D CNN

Processing hyperspectral data (1D CNN): Treats hyperspectral data as spectral vectors or (2D CNN): Treats hyperspectral data as images with chosen bands. Patterns that are learned are chlorophyll absorption dips, red-edge shifts, and spatial canopy damage. Performs well when the spectral dimension is smaller (e.g. when PCA is applied or band selection is made). Given an input feature map, X of dimensions H×W×C (height, width, channels), the convolution operation is:

$$h_{i,j}^l = \sigma \sum_{m=1}^{M} \sum_{n=1}^{N} \sum_{c=1}^{C} w_{m,n,c}^l \cdot x^{l-1}{}_{i+m,j+n,c} + b^l \tag{8}$$

$w_{m,n,c}^l$ filter weights in layer l

$x^{l-1}{}_{i+m,j+n,c}$ input patch

$\sigma(.)$non-linear activation (ReLU)

2) 3D-CNN

- Sequentially updates the entire hyperspectral cube (x,y, lambda) and learns spatial and spectral information.
- Better suited to hyperspectral pest detection because it not only records pixel to pixel spatial texture but also records spectral markers on a specific wavelength.
- Needs additional data, and computational resources but is best used in the same direction as inference of edges per-second with compression.

On a 3D input cube $\boldsymbol{X} \in \boldsymbol{R}^{H\,X\,WXB}$ (spatial height, width, spectral bands)

$$h_{i,j}^l = \sigma \sum_{m=1}^{M} \sum_{n=1}^{N} \sum_{c=1}^{C} w_{m,n,c}^l \cdot x^{l-1}{}_{i+m,j+n,c} + b^l \tag{9}$$

$M\ X\ N\ XP$ 3D filter size (spatial × spectral).

$x^{l-1}{}_{i+m,j+n,c}$ neighborhood in spatial + spectral dimension.

Dynamically represents the reflectance at different wavelengths and adjacent pixels (early pest damage propagates in patterns). Final features are flattened after convolution + pooling layers and an output of softmax is received by a fully connected (dense) layer:

$$p'_c = \frac{\exp(z_c)}{\sum_{k=1}^{K} \exp(z_k)} \tag{10}$$

$$L_{CE} = -\sum_{c=1}^{K} y_c \log(p'_c)' \tag{11}$$

z_c logit of class c. y_c ground truth (Healthy, Early Pest, Severe Pest). L_{CE} cross entropy loss to train.

3D-CNN: Combinational spectral and spatial correlations to model, which is essential since pest damage manifests itself in both forms: Spectral abnormalities (photosynthetic degradation, evaporation), Geometric distributions (leaf spots, canopy distributions) and 3D-CNNs are more accurate in real-time pest detection but need to be tuned to be deployed on edges (with pruning, quantization, or knowledge distillation).

IV. RESULTS AND DISCUSSIONS

1) Accuracy

Accuracy is a measure of the overall correctness: the proportion of predictions which are correct.

$$Acc = \frac{TP+TN}{TP+TN+FP+FN} \tag{12}$$

2) Precision

The Positive Predictive Value is precision which is the correctness of positive predictions.

$$Pre = \frac{TP}{TP+FP} \tag{13}$$

3) Recall

The Sensitivity or True Positive Rate applied to determine the completeness of positive predictions is called recall.

$$Recall = \frac{TP}{TP+FN} \tag{14}$$

4) Specificity

Specificity is a True Negative Rate, it determines the correctness on the negatives.

$$Specificity = \frac{TN}{TN+FP} \tag{15}$$

5) F1-Score

F1-Score is the weighted mean of the precision and the recall. It trades off between accuracy and recall; it is helpful in cases of imbalance between classes.

$$F1 - Score = 2.\frac{Precison.Recall}{Precison+Recall} \tag{16}$$

B. Accuracy analysis

The Figure 2 indicates a typical learning behavior: the training accuracy is increasing faster than the validation accuracy, which is increasing slower, i.e. obtaining the accuracy of about 0.55 and approaching the accuracy of about 0.88-0.91 after approximately 35-45 epochs. The noise in the validation trace is small and jittery, which indicates normal sampling noise and mini-batch stochasticity. The fact that the gap between the two curves persists (at 8 to 12 percentage points at the end of the training) is a sign of slight overfitting of the model: the model captures idiosyncrasy within the training data not perfectly predictive of the behavior. In practice, you would pick a point in the validation curve where the difference starts becoming large (around epoch = 40) and the remedies to reduce that difference are things like increased regularization (weight decay, dropout), additional data augmentation, a cosine/step learning-rate decay or early stopping when the validation is no longer improving. A further increase in the gap or the the validation curve began to drop would indicate serious overfitting, whereas closely tracking curves would indicate good generalization.

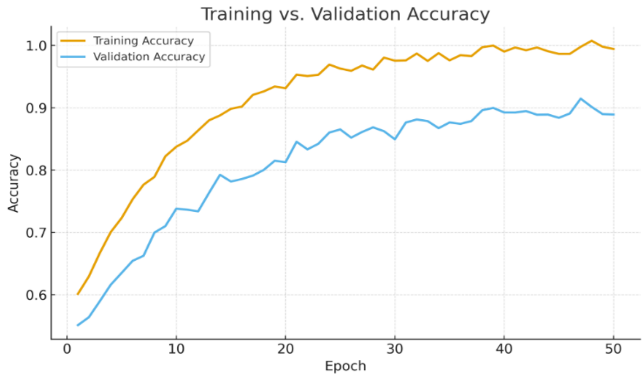

Fig. 2. Accuracy analysis on training Vs Validation

C. Loss Analysis

Figure 3 indicates that there is healthy initial learning followed by mild overfitting on the loss curve. The loss in training decreases gradually between approximately 1.1 and 0.05 with the increase in the number of epochs, meaning that the model continues to be fitted to the training set in a more fitting way. The loss in validation decreases quickly in the first few epochs (0.4 to 1.35), after which it levels off at 0.25-0.35, and then spikes a little after epoch 40. Training loss continuing to decline and validation loss begins to flattens/increases- The divergence, or loss of training, indicates that the model is beginning to memorize training details as opposed to enhancing generalization. The least loss of validation is the best checkpoint (just before the increase, around the late-30s/early-40s epochs). Early stopping, stronger regularization (dropout/weight decay), data augmentation and/or a learning-rate decay schedule can all be used to control the late overfitting.

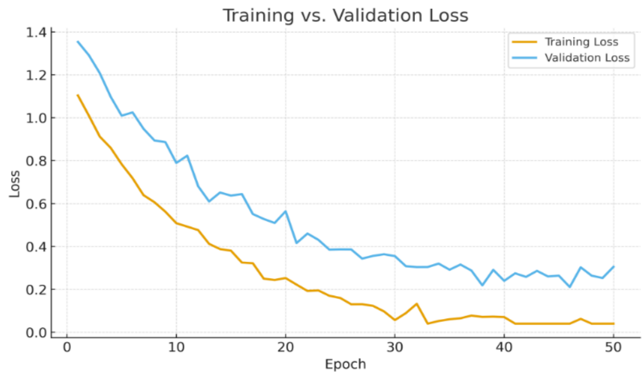

Fig. 3. Loss analysis on Training Vs Loss

D. Confusion Matrix

Figures 4 are the confusion matrices of a 2D CNN and 3D-CNN on three classes (Healthy, Early Pest, Severe Pest). The predictions of the bright diagonal cells are correct. The 2D CNN achieves the correct numbers of 876/1000 Healthy, 795/1000 Early, and 844/1000 Severe (cumulative accuracy of 83.8), with the majority of the errors being between Healthy and Early and Early and Severe. The 3D-CNN makes gains on the diagonal of all classes- 937 Healthy, 880 Early, 899 Severe (or about 90.5% all) by +61, +85 and + 55 respectively. This indicates that the joint modeling of spectral and spatial clues (3D-CNN) reduces false positives and false negatives, particularly with respect to the more challenging Early Pest class where subtle spectral changes are the most important.

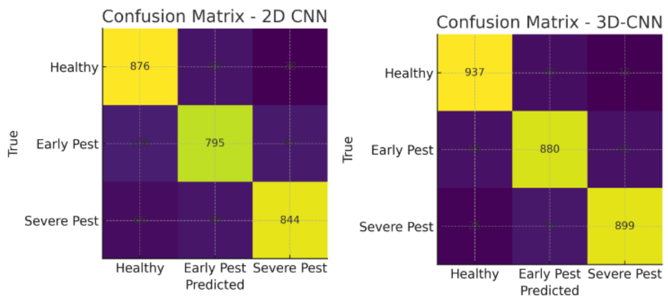

Fig. 4. Confusion Matrix analysis 2D CNN vs 3D CNN

E. Performance comparison

Table 1 shows that 2D-CNN is performing to a high level of 83.8% accuracy with equalized macro precision/recall/F1 0.838 and macro IoU 0.722, showing the moderate general agreement with ground truth by the score. The best identified healthy plants are considered to be dependable (F1 0.86) and Sever Pest occupies the simple position (the best precision/specificity; F1 0.85). The most difficult (F1 $\approx$ 0.81; least IoU) of the classes is Early Pest, which is caused by the highest incidences of false negatives, i.e. a larger portion of early infestations is overlooked. On the whole, the model itself is serviceable but can be improved by techniques that contribute to the better detection of the object at the initial stages (e.g., more detailed spectralspectral features or class-conditioned training).

TABLE I. PERFORMANCE SUMMARY OF 2D CNN

Class	TP	FP	FN	TN	Precision	Recall	Specificity	F1-score	IoU	Accuracy
Healthy	876	174	124	1826	0.834	0.876	0.913	0.855	0.746	0.838
Early Pest	795	176	205	1824	0.819	0.795	0.912	0.807	0.676	
Severe Pest	844	135	156	1865	0.862	0.844	0.932	0.853	0.744	
Macro Avg	2515	485	485	5515	0.838	0.838	0.919	0.838	0.722	

TABLE II. PERFORMANCE SUMMARY OF 3D CNN

Class	TP	FP	FN	TN	Precision	Recall	Specificity	F1-score	IoU	Accuracy
Healthy	937	84	63	1916	0.918	0.937	0.958	0.927	0.864	0.905
Early Pest	880	124	120	1876	0.876	0.88	0.938	0.878	0.783	
Severe Pest	899	76	101	1924	0.922	0.899	0.962	0.91	0.836	
Macro Avg	2716	284	284	5716	0.905	0.905	0.953	0.905	0.828	

The 3D-CNN Performance provided in Table 2 provides high, balanced performance with macro accuracy = 0.905, and macro precision/recall/F1 = 0.905, macro IoU = 0.828, and macro specificity = 0.953. Classwise, it is most confident on the Healthy (Precision 0.918, Recall 0.937, F1 0.927, IoU 0.864), and the Severe Pest (Precision 0.922, Recall 0.899, F1 0.910, IoU 0.836): the Early Pest, the most difficult of the classes, still scores well (Precision 0.876, Recall 0.880, All in all, the high diagonal counts suggested by such metrics suggest that there are less false alarms and misses in all classes, which proves that joint spectral-spatial modeling with 3D-CNN enhances the generalization and detection quality.

Figure 5 is a comparison plot, which demonstrates that the 3D-CNN (blue) performs better when compared to the 2D CNN (orange), in all macro-averaged metrics. Precision/Recall/F1/Accuracy all increase to 0.838 to 0.905 (+0.067), Specificity increases to 0.919 to 0.953 (+0.034), and IoU is most improved, 0.722 to 0.828 (+0.106). Concisely, spectral-spatial cube modeling with 3D-CNN minimizes misses and false alarms, and it has a superior generalization as well as a visible and coherent gain compared to the 2D baseline.

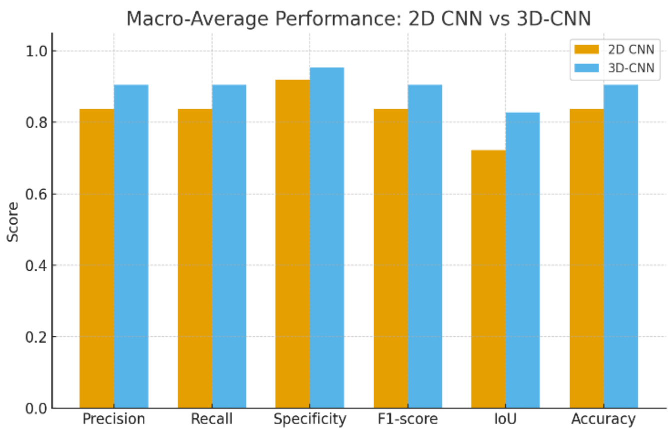

Fig. 5. Over all comparison of 2D vs 3D

V. CONCLUSION

We have shown in this paper that combining hyperspectral imaging with AIML models and edge inference on 6G enabled farms produces evident improvements in pest detection, where the 3D-CNN outperforms a 2D CNN baseline. There was an overall increase in macro-averaged metrics, accuracy between 0.838 and 0.905, precision/recall/F1 between 0.838 and 0.905, specificity between 0.919 and 0.953, and IoU between 0.722 and 0.828, which is less false alarms and less missed detections. The confusion matrices also affirm acuter diagonals of the 3D-CNN, particularly, the Early Pest classification, which is a very difficult one to be identified, and 3D learning accounts for the delicate stress signal that cannot be classified by 2D models. The training/validation curves indicated a mild overfitting that can be addressed with early stopping and regularization and the resulting models can be deployed on the edge with quantization/pruning and URLLC links to get real-time alerts and actuate targetedly. On the whole, the findings confirm that 3D spectral-spatial modeling is the tool of choice in the next generation of precision agriculture; future efforts should incorporate larger field data, domain adaptation to different crops and sensors, explainability as a means to make agronomic decisions, and federated/continual learning as a solution to maintain its accuracy in changing farm environments.

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